Supporting information

Relationship between iron carbide phases (ϵ-Fe_2C, Fe_7C_3, and χ-Fe_5C_2) and catalytic performances of Fe/SiO_2 Fischer-Tropsch catalyst

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Table S1 XRF results of the calcined catalyst.

No.	Component	Content (mass %)	Measured elemental line
1	Fe_2O_3	72.6	Fe-Kα
2	SiO_2	27.3	Si-Kα (FA)
3	Na_2O	0.0010	Na-Kα (FA)
4	MgO	0.0090	Mg-Kα (FA)
5	Al_2O_3	0.0132	Al-Kα (FA)
6	P_2O_5	0.0056	P-Kα
7	SO_3	0.0022	S-Kα
8	Cl	0.0029	Cl-Kα (FA)
9	K_2O	0.0057	K-Kα
10	CaO	0.0167	Ca-Kα
11	Cr_2O_3	0.0196	Cr-Kα
12	MnO	0.0180	Mn-Kα
13	NiO	0.0195	Ni-Kα
14	CuO	0.0148	Cu-Kα
15	ZnO	0.0120	Zn-Kα
16	ZrO_2	0.0066	Zr-Kα

Table S2 ICP-AES results of the calcined catalyst.

Element	Content
Fe	50.7%
Si	12.7%

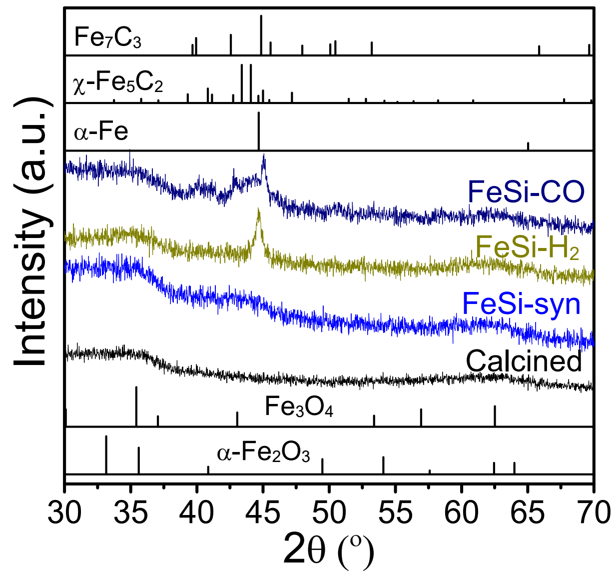

Figure S1 XRD patterns of the reduced catalysts after different atmospheres at 300 ºC for 20 h.

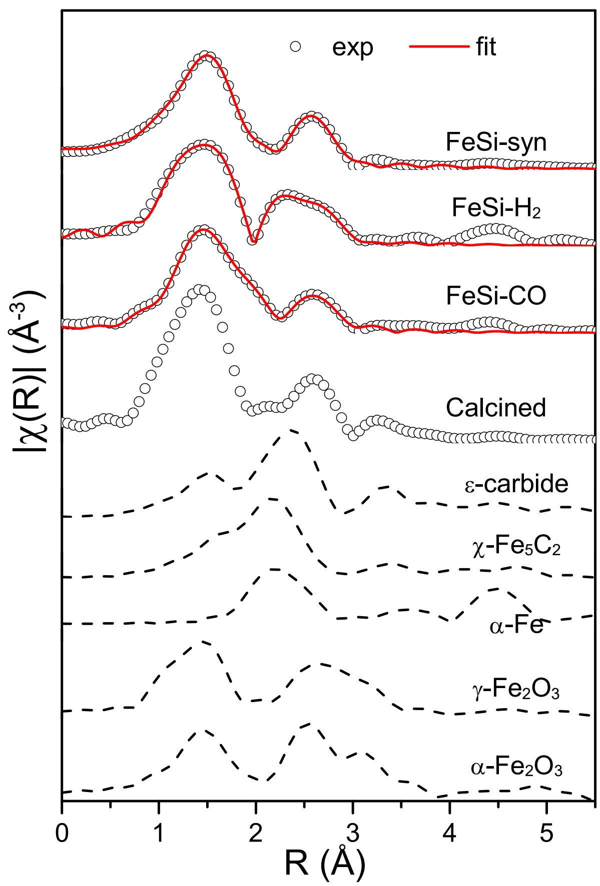

Figure S2 Fourier-transformed EXAFS spectra of the reduced catalysts after different atmospheres at 300 °C for 20 h.

Table S3 Results of EXAFS fitting of the reduced catalysts after different atmospheres at 300 °C for 20 h.

Catalysts	Shell	N	R (Å)	σ^2 (Å2)	ΔE (eV)
FeSi-H_2	Fe-O	4.9	1.72-2.29	0.020	2.9
	Fe-O	2.2	3.24	0.020	-3.2
	Fe-Fe	11.1	2.31-3.38	0.020	-3.2
FeSi-CO	Fe-O	1.5	1.93-2.10	0.0028	9.1
	Fe-Fe	0.9	3.03	0.0028	0.2
	Fe-C	1.3	1.80	0.0039	-9.1
	Fe-Fe	2.5	2.20-2.42	0.0039	-9.1
FeSi-syn	Fe-O	5	1.99	0.013	1.4
	Fe-Fe	7.5	3.12	0.020	7.0
	Fe-C	0.1	2.15	0.0080	11.0
	Fe-Fe	0.2	2.54-2.72	0.0080	11.0

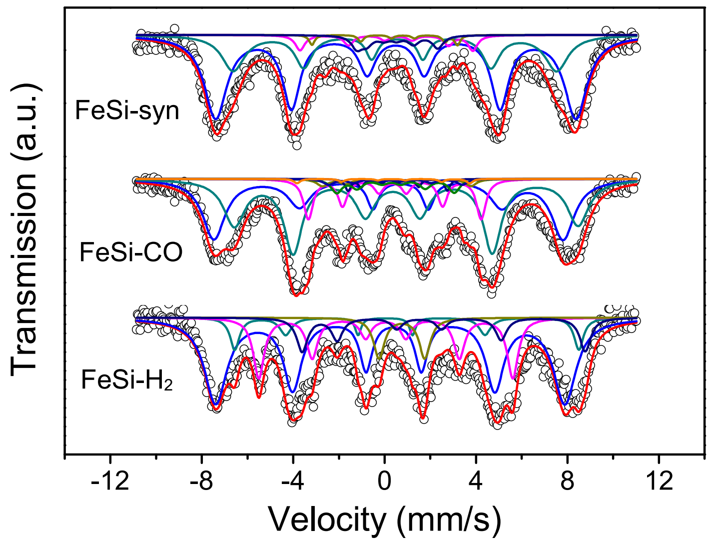

Figure S3 MES spectra of the reduced catalysts measured at 11 K.

Table S4 Iron phase compositions of the reduced catalysts from MES at 11 K.

Reduced catalysts	IS (mm/s)	QS (mm/s)	Hhf (kOe)	Phase	Compositions (%)
FeSi-H_2	0.32		475	Fe_3O_4 (A)	
	0.50		468	Fe_3O_4 (B1)	74.6
	1.20		385	Fe_3O_4 (B2)	
	0.75	2.0		Fe^{2+}	6.4
	0.05		345	α-Fe	19.0
FeSi-CO	0.40		475	Fe_3O_4 (A)	64.9
	0.65		464	Fe_3O_4 (B)	
	0.16		225	χ-Fe_5C_2 (I)	
	0.00		187	χ-Fe_5C_2 (II)	19.7
	0.16		130	χ-Fe_5C_2 (III)	
	0.32		163	Fe_7C_3 (I)	
	0.40		190	Fe_7C_3 (II)	15.4
	0.15		208	Fe_7C_3 (III)	
	0.12		229	Fe_7C_3 (IV)	
FeSi-syn	0.49		490	Fe_3O_4 (A)	83.9
	0.50		440	Fe_3O_4 (B)	
	0.34		235	χ-Fe_5C_2 (I)	
	0.40		198	χ-Fe_5C_2 (II)	16.1
	0.40		110	χ-Fe_5C_2 (III)	

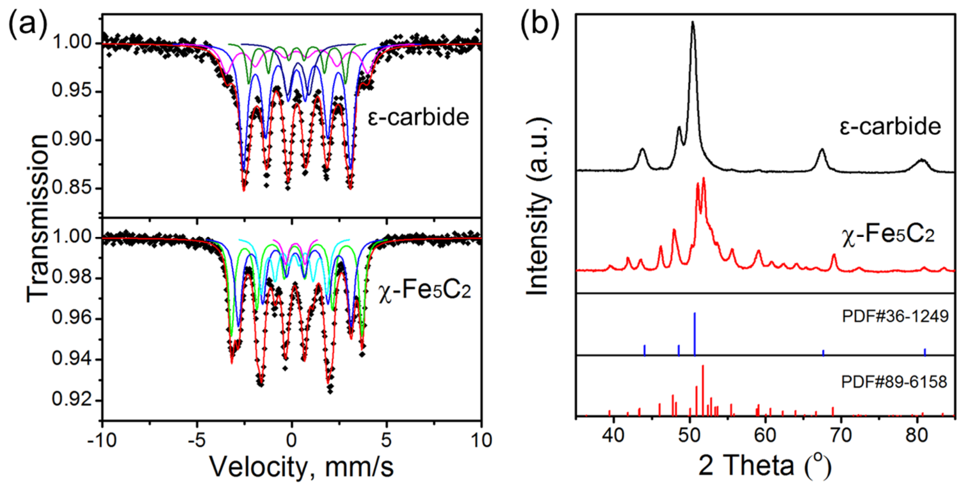

Figure S4 MES and XRD spectra of ϵ-carbide and χ-Fe_5C_2 carbide standards. ϵ-carbide and χ-Fe_5C_2 standard carbides were prepared using the gas carburization method. First, the precursor α-Fe_2O_3 was reduced to α-Fe by H_2 (99.999%, 80 ml/min) at 300 °C for 24 h. The sample was carburized by 1:4 CO/H_2 at 180 °C for 48 h to synthesize ε-carbide, and χ-Fe_5C_2 was acquired by annealing ε-carbide under Ar (99.999%, 50 ml/min) at 350 °C.

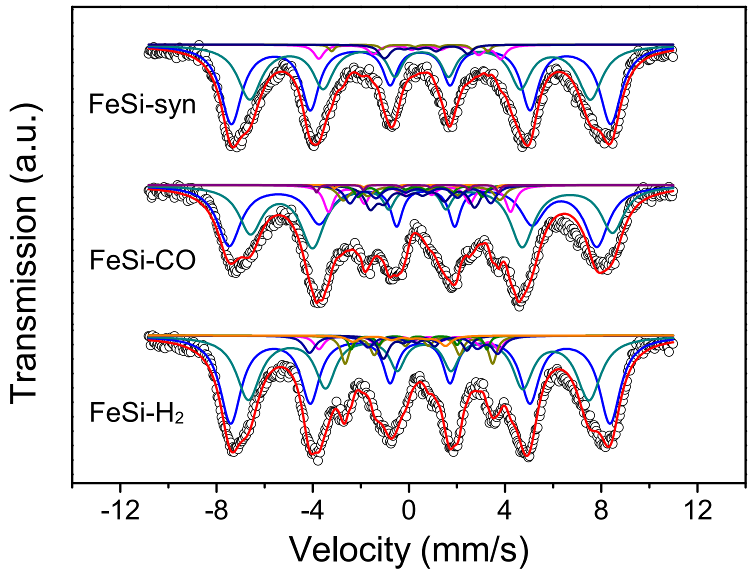

Figure S5 MES spectra of the spent catalysts measured at 11 K.

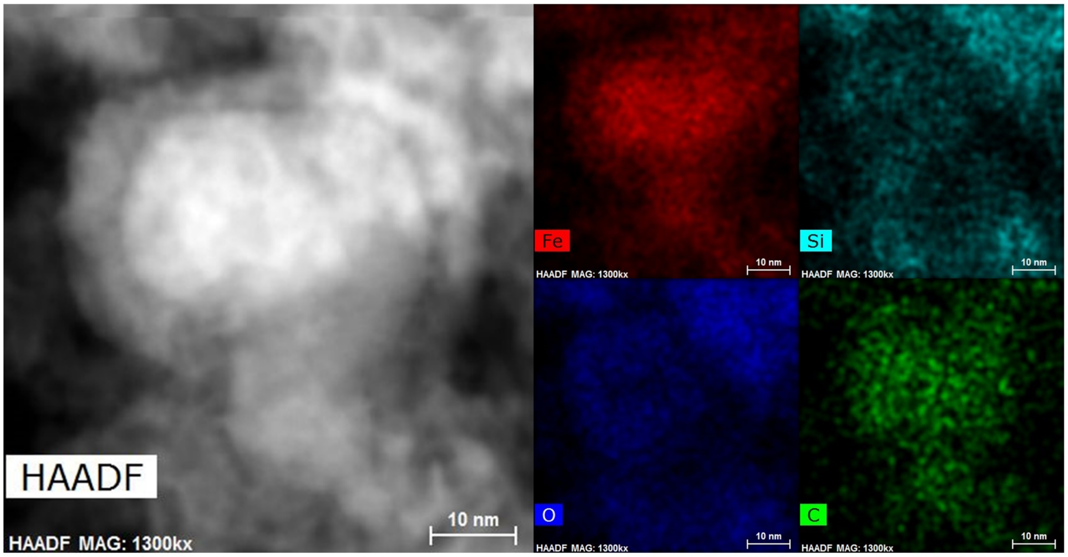

Figure S6 STEM elementary mapping of the spent FeSi-CO catalyst.

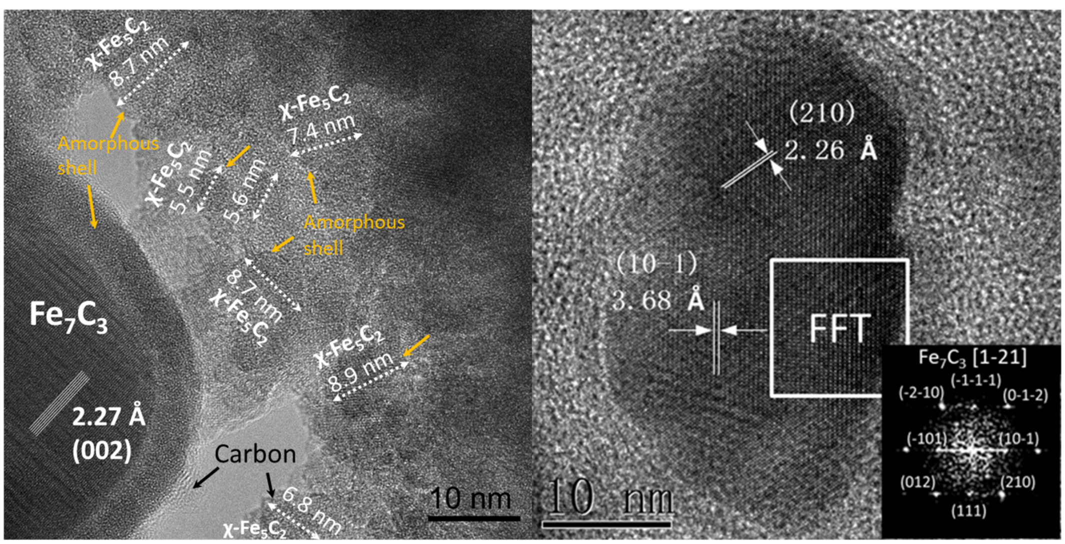

Figure S7 TEM of the spent FeSi-CO catalyst.

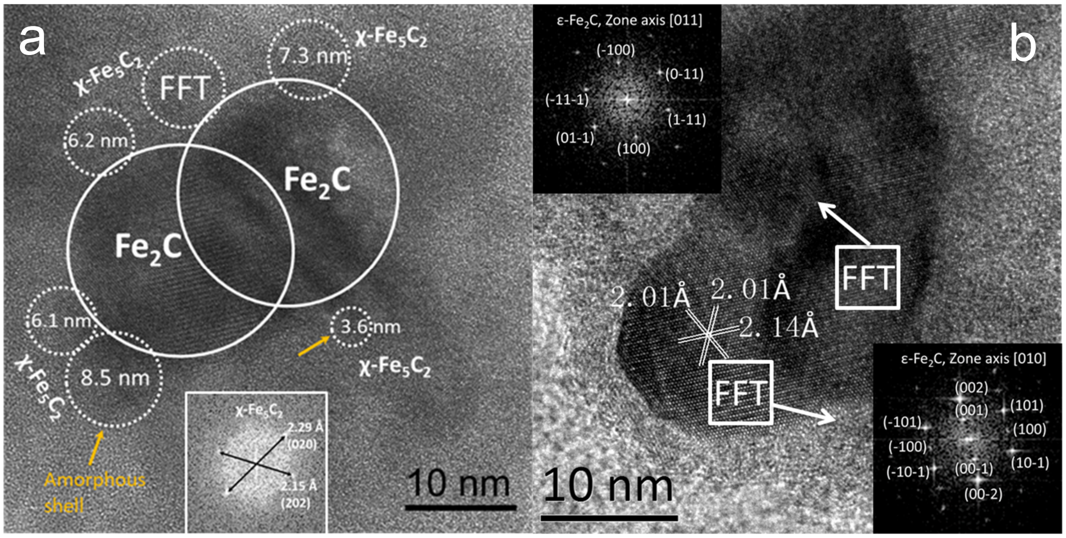

Figure S8 TEM of the spent FeSi-H_2 catalyst.

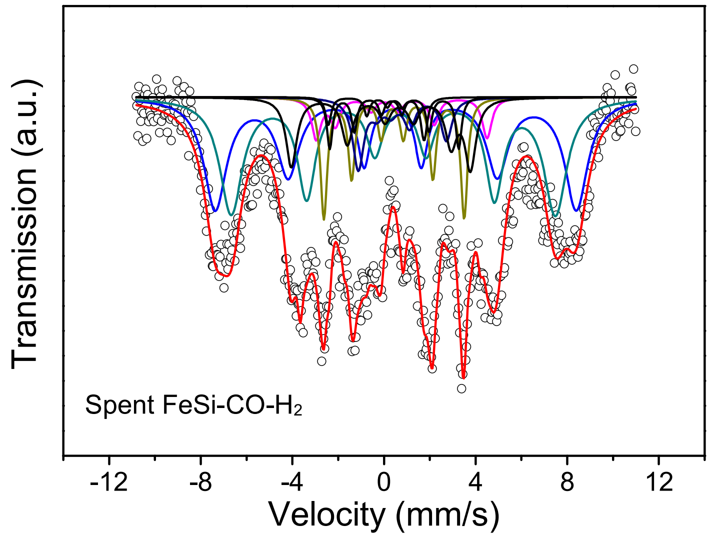

Figure S9 MES spectra of the spent FeSi-CO-H_2 catalyst measured at 11 K.

Table S5 Iron phase compositions of the spent FeSi-CO-H_2 catalyst from MES at 11 K.

Spent catalyst	IS (mm/s)	Hhf (kOe)	Phase	Compositions (%)
FeSi-CO-H_2	0.45	490	Fe_3O_4 (A)	59.6
	0.56	440	Fe_3O_4 (B)	
	0.40	232	χ-Fe_5C_2 (I)	
	0.40	190	χ-Fe_5C_2 (II)	15.3
	0.40	120	χ-Fe_5C_2 (III)	
	0.39	175	ε-Fe_2C (I)	
	0.27	243	ε-Fe_2C (II)	22.7
	0.07	131	ε-Fe_2C (III)	

Table S6 Comparison of TOFs between this work and literature.

Sample	Reaction conditions	Iron phase	Content in iron	TOF (s^{-1})	Reference
FeSi	260 ºC, 3.0 MPa, H_2/CO=2	χ-Fe_5C_2	11.0-16.5%	0.0162	This work
FeSi	260 ºC, 3.0 MPa, H_2/CO=2	ϵ-Fe_2C	14.7-22.7%	0.0150	This work
FeSi	260 ºC, 3.0 MPa, H_2/CO=2	Fe_7C_3	16.3%	0.0459	This work
Fe/xSiO_2	280 ºC, 1.5 MPa, H_2/CO=2	χ-Fe_5C_2	2.3-37.3%	0.038	33
Fe/SBA-15	430 ºC, 0.1 MPa, H_2/CO=2	χ-Fe_5C_2	12%	0.013	63
Fe/CNF	350 ºC, 0.1 MPa, H_2/CO=2	χ-Fe_5C_2	50-80%	0.002-0.012	61
Fe/CNF	340 ºC, 2.0 MPa, H_2/CO=1	χ-Fe_5C_2	~80%	0.01-0.08	61
Fe@C	340 ºC, 2.0 MPa, H_2/CO=1	χ-Fe_5C_2	~90%	0.07-0.11	66
RQ Fe	150 ºC, 3 MPa, H_2/CO=2	ϵ-Fe_2C	73%	0.03	12
Fe@C-450	340 ºC, 2.0 MPa, H_2/CO=0.5	ϵ-$Fe_{2.2}C$	53%	0.06	67